Akoizo Original Practice - Form 3

Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A ramp is 30 cm long and rises 12 cm. What is its IMA?

A. 2.5

B. 0.4

C. 3.6

D. 18

Answer: _____

2. (2 points) A device has one dimension of 83 cm. What happens?

A. It runs but is scored in Tier 2, with a 25-point penalty

B. It cannot run

C. Only a 25-point penalty

D. It is scored in Tier 3

Answer: _____

3. (2 points) Why does a water timer slow down as it drains?

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

A. The hole gets smaller

B. Air pressure rises

C. Outflow speed depends on the water height above the hole

D. Water gets thicker

Answer: _____

4. (2 points) Why might Alka-Seltzer timing differ at competition?

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

A. The coin order changes it

B. Gravity is different

C. Reaction speed depends on water temperature

D. Tablets are a different color

Answer: _____

5. (4 points) Explain why a selective coin trigger can be worth 175 points.

Answer: _____
_____
_____

6. (4 points) Name the three ASL columns and what each contains.

Unit: The Final Action, the ASL, and competition day.
Worked-case data: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.
Separate investigation observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

Answer: _____
_____
_____

7. (4 points) How can a falling magnetic object be released by a previous action?

Answer: _____
_____
_____

8. (4 points) Describe a gravity-driven screw action.

Unit: Simple machines: levers, inclines, pulleys, and screws.
Worked-case data: A screw has a pitch of 1.5 mm. How many turns move an object 2.0 cm, and is that enough rotation for the action?
Separate investigation observations:
Pulley build: Load 420 g lifts 11 cm; effort rope moves 22 cm.
Supervisor check: Counts two supporting segments.
Team claim: Pulley action meets IMA 3.

Answer: _____
_____
_____

9. (4 points) What happens to scoring if a team touches the device for a fourth time?

Answer: _____
_____
_____

10. (4 points) What happens to time points if the run reaches twice the Target Time?

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?

Answer: _____
_____
_____

12. (2 points) Which gap should the team use?

Supplied synthetic observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

A. 21.5 mm

B. 24.0 mm

C. 22.8 mm

Answer: _____

13. (2 points) Why did the change help?

Supplied synthetic observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

A. The platform got walls

B. The pack fell flat instead of tumbling, so it did not bounce off

C. The drop became much shorter

Answer: _____

14. (2 points) Define “IR LED” in the context of Mission Possible.

Answer: _____
_____
_____

15. (2 points) Define “Fulcrum” in the context of Mission Possible.

Answer: _____
_____
_____

16. (2 points) Define “Quarter-only start” in the context of Mission Possible.

Answer: _____
_____
_____

17. (2 points) Define “ASL columns” in the context of Mission Possible.

Answer: _____
_____
_____

18. (2 points) Define “Flow rate” in the context of Mission Possible.

Answer: _____
_____
_____